

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #09 — Faculty Research Grant & Publication Tracker

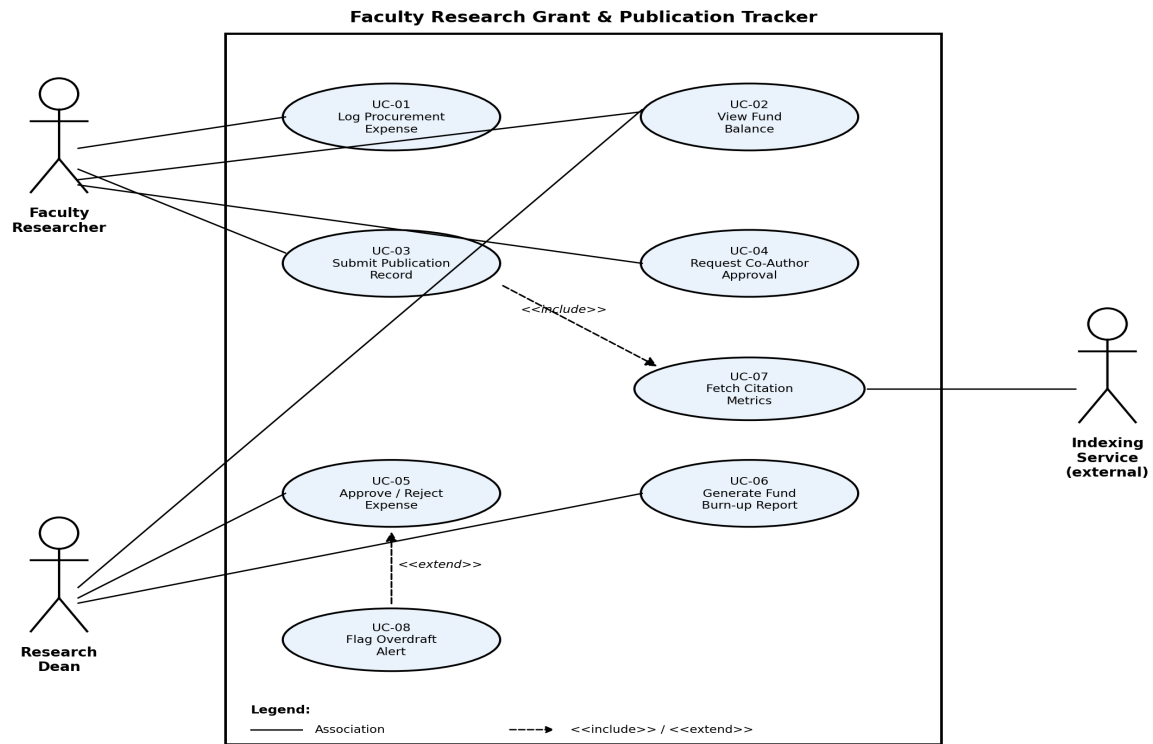
Stakeholders / Actors: Faculty Researcher, Research Dean, Indexing Service (external)

1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall track grant fund allocations, allow faculty to log equipment procurement expenses, and compute remaining budgetary balance.	High	Pass: Balance updates accurately and blocks expenses exceeding grant cap. Fail: Overdraft allowed without admin override.	Core financial control needed to prevent unauthorised over-spending of sponsored grants.
FR-002	Functional	The system shall allow a faculty researcher to submit a publication record (title, DOI, venue) and automatically fetch its indexing status and citation count from the linked indexing service.	High	Pass: A valid DOI returns indexing status and citation count within the same session and stores it against the faculty profile. Fail: Record is saved with no indexing/citation data and no error shown.	Publication metrics are the primary output metric the research deanery reports to funding bodies.
FR-003	Functional	The system shall require explicit approval from every listed co-author before a submitted publication record is marked as finalised.	Medium	Pass: Record status remains 'Pending Co-Author Approval' until all co-authors approve; status changes to 'Finalised' only when the last approval is received. Fail: Record finalises with one or more co-authors not having responded.	Prevents disputes over authorship credit and ensures accurate multi-author attribution.
FR-004	Functional	The system shall allow the Research Dean to approve or reject any logged expense that exceeds a configurable percentage of the remaining grant balance.	High	Pass: Expenses above the configured threshold are held in 'Awaiting Dean Approval' status and are only posted to the ledger after explicit approval. Fail: A threshold-exceeding expense is posted without Dean action.	Adds a human control point for high-value spend, complementing the automated cap check in FR-001.
FR-005	Functional	The system shall generate a fund burn-up report per grant, showing cumulative expenditure against the allocated budget over a selected date range, exportable as PDF or CSV.	Medium	Pass: Report totals reconcile exactly with the sum of approved ledger entries for the selected range and export completes in a valid PDF/CSV file. Fail: Report totals do not match the ledger or export fails silently.	Gives the Research Dean the analytics needed for grant renewal and utilisation reviews.
NFR-001	Nonfunctional	The system shall maintain an immutable, tamper-evident audit ledger for all financial approvals and journal status modifications.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load; any post-hoc edit attempt is detectable via hash-chain verification. Fail: A ledger entry can be altered without detection.	Type: Performance & Security. Sponsored-research funds are subject to audit; records must be legally defensible.
NFR-002	Nonfunctional	The system shall synchronise citation and indexing metrics from the external indexing service within 5 seconds under normal load, and shall serve the last cached values with a 'stale data' indicator if the service is unreachable.	Medium	Pass: 95% of sync requests complete within 5 seconds in load testing, and cached data with a visible staleness flag is served during a simulated indexing-service outage. Fail: The UI blocks or errors out when the indexing service is unreachable.	Type: Performance & Reliability. Third-party indexing APIs are outside the system's control and must not block core workflows.

2. UML Use-Case Diagram

Actors: **Faculty Researcher**, **Research Dean** (primary/human actors) and **Indexing Service** (external system actor providing citation/indexing data). UC-03 «includes» UC-07 because every publication submission must always retrieve citation metrics. UC-08 «extends» UC-05, since the overdraft-flag behaviour is optional and only triggers when the logged expense would exceed the grant cap.



3. Use-Case Flow Specification

Use Case: UC-01 — Log Procurement Expense

(Corresponds to FR-001, the primary financial-control requirement of the system.)

Primary Actor

Faculty Researcher

Preconditions

- The Faculty Researcher is authenticated and is listed as an investigator on an active, approved grant.
- The grant has a valid, non-zero allocated budget on record.

Postconditions

- Success: A new expense entry is written to the immutable audit ledger and the grant's remaining balance is updated accordingly.
- Failure: No ledger entry is created and the previous balance remains unchanged.

Main Success Scenario

- 1 Faculty Researcher selects "Log Expense" from the grant dashboard.
- 2 System displays an expense-entry form (category, vendor, amount, date, supporting invoice upload).
- 3 Faculty Researcher enters the expense details and uploads the invoice, then submits the form.
- 4 System validates the input (mandatory fields complete, amount is numeric and positive, invoice file attached).
- 5 System checks the expense amount against the grant's remaining budgetary balance.
- 6 System confirms the amount is within the grant cap.
- 7 System writes the expense as a new entry in the tamper-evident audit ledger and recomputes the remaining balance.
- 8 System displays a confirmation message with the updated balance and a receipt reference number.
- 9 Use case ends successfully.

Alternate Flow

6a. Expense Exceeds Grant Cap

- 6a1. System detects that the submitted amount would exceed the remaining grant balance.
- 6a2. System blocks automatic posting and flags the expense as "Awaiting Dean Approval" (triggers UC-08 «extend» → UC-05).
- 6a3. System notifies the Research Dean that an over-cap expense is pending review.
- 6a4. If the Research Dean approves the override, the system posts the entry to the ledger and updates the balance; if rejected, the entry is discarded and the Faculty Researcher is notified with the reason.